

Elasticsearch - Statistical analysis and advanced uses

Download the dataset

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ wget https://dst-de.s3.eu-west-3.amazonaws.com/elasticsearch_fr/datasets/cars.csv
--2025-07-24 17:18:08-- https://dst-de.s3.eu-west-3.amazonaws.com/elasticsearch_fr/datasets/cars.csv
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 16.12.19.6, 52.95.156.82
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|16.12.19.6|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2928524 (2.8M) [text/csv]
Saving to: 'cars.csv'

cars.csv                               100%[=====] 2.79M  14.3MB/s   in 0.2s

2025-07-24 17:18:08 (14.3 MB/s) - 'cars.csv' saved [2928524/2928524]
```

Create file second_bulk_script.py to import the cars.csv data in Elasticsearch

Run the script

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ python3 second_bulk_script.py
/usr/lib/python3/dist-packages/requests/__init__.py:89: RequestsDependencyWarning: urllib3 (1.26.20) or char
det (3.0.4) doesn't match a supported version!
warnings.warn("urllib3 ({}), or chardet ({}), doesn't match a supported "
```

Verify

The screenshot shows the Elasticsearch console interface with two REST API requests and their corresponding JSON responses.

Request 1: GET /cars/_mapping

Response 1: A JSON object describing the mappings for the 'cars' index. It includes fields like 'color', 'fuel_type', 'make', and 'mileage(kilometers)', each with a 'keyword' type and 'ignore_above': 256.

Request 2: GET /cars/_search with a query: {"query": {"match_all": {}}}

Response 2: A JSON object showing search results. It includes a 'took' value of 453, a 'total' of 1, and a 'hits' array containing one document with fields like 'make', 'priceUSD', 'year', 'mileage(kilometers)', 'fuel_type', 'volume(cn3)', 'color', 'transmission', and 'segment'.

Reindex into properly typed index and alias

Create new index with cars_new with explicit mappings

ConsoleSearch ProfilerGrok DebuggerPainless LabBETA

HistorySettingsHelp

1-200 - OK800 ms

```
1 GET /cars/_mapping
2
3 GET /cars/_search
4 {
5   "query": { "match_all": {} }
6 }
7
8 PUT /cars_new
9 {
10  "mappings": {
11    "properties": {
12      "color": { "type": "text", "fields": { "keyword"
13        : { "type": "keyword" } } },
14      "fuel_type": { "type": "text", "fields": { "keyword"
15        : { "type": "keyword" } } },
16      "make": { "type": "text", "fields": { "keyword"
17        : { "type": "keyword" } } },
18      "mileage(kilometers)": { "type": "float" },
19      "priceUSD": { "type": "float" },
20      "segment": { "type": "text", "fields": { "keyword"
21        : { "type": "keyword" } } },
22      "transmission": { "type": "text", "fields": { "keyword"
23        : { "type": "keyword" } } },
24      "volume(cm3)": { "type": "float" },
25      "year": { "type": "date", "format": "yyyy" }
26    }
27  }
28 }
```

```
1 {
2   "acknowledged" : true,
3   "shards_acknowledged" : true,
4   "index" : "cars_new"
5 }
```

Click to send request

Reindex the old data into cars_new

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HistorySettingsHelp

1-200 - OK17692 ms

```
1 GET /cars/_mapping
2
3 GET /cars/_search
4 {
5   "query": { "match_all": {} }
6 }
7
8 PUT /cars_new
9 {
10  "mappings": {
11    "properties": {
12      "color": { "type": "text", "fields": { "keyword"
13        : { "type": "keyword" } } },
14      "fuel_type": { "type": "text", "fields": { "keyword"
15        : { "type": "keyword" } } },
16      "make": { "type": "text", "fields": { "keyword"
17        : { "type": "keyword" } } },
18      "mileage(kilometers)": { "type": "float" },
19      "priceUSD": { "type": "float" },
20      "segment": { "type": "text", "fields": { "keyword"
21        : { "type": "keyword" } } },
22      "transmission": { "type": "text", "fields": { "keyword"
23        : { "type": "keyword" } } },
24      "volume(cm3)": { "type": "float" },
25      "year": { "type": "date", "format": "yyyy" }
26    }
27  }
28 }
```

```
1 {
2   "took" : 17399,
3   "timed_out" : false,
4   "total" : 50849,
5   "updated" : 0,
6   "created" : 50849,
7   "deleted" : 0,
8   "batches" : 51,
9   "version_conflicts" : 0,
10  "noops" : 0,
11  "retries" : {
12    "bulk" : 0,
13    "search" : 0
14  },
15  "throttled_millis" : 0,
16  "requests_per_second" : -1.0,
17  "throttled_until_millis" : 0,
18  "failures" : [ ]
19 }
```

Click to send request

Delete the original

ConsoleSearch ProfilerGrok DebuggerPainless LabBETA

HistorySettingsHelp

1-200 - OK446 ms

```
1 GET /cars/_mapping
2
3 GET /cars/_search
4 {
5   "query": { "match_all": {} }
6 }
7
8 PUT /cars_new
9 {
10  "mappings": {
11    "properties": {
12      "color": { "type": "text", "fields": { "keyword"
13        : { "type": "keyword" } } },
14      "fuel_type": { "type": "text", "fields": { "keyword"
15        : { "type": "keyword" } } },
16      "make": { "type": "text", "fields": { "keyword"
17        : { "type": "keyword" } } },
18      "mileage(kilometers)": { "type": "float" },
19      "priceUSD": { "type": "float" },
20      "segment": { "type": "text", "fields": { "keyword"
21        : { "type": "keyword" } } },
22      "transmission": { "type": "text", "fields": { "keyword"
23        : { "type": "keyword" } } },
24      "volume(cm3)": { "type": "float" },
25      "year": { "type": "date", "format": "yyyy" }
26    }
27  }
28 }
```

```
1 {
2   "acknowledged" : true
3 }
```

Click to send request

Alias the new one as cars

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HistorySettingsHelp200 - OK381 ms

```
6- }
7
8 PUT /cars_new
9 {
10-   "mappings": {
11-     "properties": {
12-       | "color": { "type": "text", "fields": { "keyword"
13-         : { "type": "keyword" } } },
14-       | "fuel_type": { "type": "text", "fields": { "keyword"
15-         : { "type": "keyword" } } },
16-       | "make": { "type": "text", "fields": { "keyword"
17-         : { "type": "keyword" } } },
18-       | "mileage(kilometers)": { "type": "float" },
19-       | "priceUSD": { "type": "float" },
20-       | "segment": { "type": "text", "fields": { "keyword"
21-         : { "type": "keyword" } } },
22-       | "transmission": { "type": "text", "fields": { "keyword"
23-         : { "type": "keyword" } } },
24-       | "volume(cm3)": { "type": "float" },
25-       | "year": { "type": "date", "format": "yyyy" }
26-     }
27-   }
28- }
29
30 POST /_reindex
31 {
32-   "source": { "index": "cars" },
33-   "dest": { "index": "cars_new" }
34- }
35
36 DELETE /cars
37
38 POST /_aliases
39 {
40-   "actions": [
41-     { "add": { "index": "cars_new", "alias": "cars" } }
42-   ]
43- }
```

```
1- {
2-   "acknowledged" : true
3- }
4
```

Click to send request

Aggregations: Basic Stats

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HistorySettingsHelp200 - OK258 ms

```
15-   | "mileage(kilometers)": { "type": "float" },
16-   | "priceUSD": { "type": "float" },
17-   | "segment": { "type": "text", "fields": { "keyword"
18-     : { "type": "keyword" } } },
19-   | "transmission": { "type": "text", "fields": { "keyword"
20-     : { "type": "keyword" } } },
21-   | "volume(cm3)": { "type": "float" },
22-   | "year": { "type": "date", "format": "yyyy" }
23- }
24
25 POST /_reindex
26 {
27-   "source": { "index": "cars" },
28-   "dest": { "index": "cars_new" }
29- }
30
31 DELETE /cars
32
33 POST /_aliases
34 {
35-   "actions": [
36-     { "add": { "index": "cars_new", "alias": "cars" } }
37-   ]
38- }
39
40 GET /cars/_search
41 {
42-   "size": 0,
43-   "aggs": {
44-     | "total_price": { "sum": { "field": "priceUSD" } },
45-     | "avg_price": { "avg": { "field": "priceUSD" } },
46-     | "min_price": { "min": { "field": "priceUSD" } },
47-     | "max_price": { "max": { "field": "priceUSD" } }
48-   }
49- }
```

```
1- {
2-   "took": 189,
3-   "timed_out" : false,
4-   "_shards": {
5-     | "total": 1,
6-     | "successful": 1,
7-     | "skipped": 0,
8-     | "failed": 0
9-   },
10-   "hits": {
11-     | "total": {
12-       | "value": 10000,
13-       | "relation": "gte"
14-     },
15-     | "max_score": null,
16-     | "hits": [ ]
17-   },
18-   "aggregations": {
19-     | "max_price": {
20-       | "value": 170000.0
21-     },
22-     | "min_price": {
23-       | "value": 48.0
24-     },
25-     | "total_price": {
26-       | "value": 3.69493993E8
27-     },
28-     | "avg_price": {
29-       | "value": 7266.494778658381
30-     }
31-   }
32- }
33
```

Click to send request

Condensed Stats and Counts

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History Settings Help 200 - OK 390 ms

```
24 POST /_reindex
25 {
26   "source": { "index": "cars" },
27   "dest": { "index": "cars_new" }
28 }
29 }
30 DELETE /cars
31
32 POST /_aliases
33 {
34   "actions": [
35     { "add": { "index": "cars_new", "alias": "cars" } }
36   ]
37 }
38 }
39 GET /cars/_search
40 {
41   "size": 0,
42   "aggs": {
43     "total_price": { "sum": { "field": "priceUSD" } },
44     "avg_price": { "avg": { "field": "priceUSD" } },
45     "min_price": { "min": { "field": "priceUSD" } },
46     "max_price": { "max": { "field": "priceUSD" } }
47   }
48 }
49 }
50
51 GET /cars/_search
52 {
53   "size": 0,
54   "aggs": {
55     "value_count": { "value_count": { "field": "priceUSD" } },
56     "price_stats": { "stats": { "field": "priceUSD" } },
57     "extended_stats": {
58       "extended_stats": { "field": "priceUSD" }
59     }
60   }
61 }
62 }
```

Click to send request

```
1- {
2   "took": 129,
3   "timed_out": false,
4   "_shards": {
5     "total": 1,
6     "successful": 1,
7     "skipped": 0,
8     "failed": 0
9   },
10  "hits": {
11    "total": {
12      "value": 10000,
13      "relation": "gte"
14    },
15    "max_score": null,
16    "hits": [ ]
17  },
18  "aggregations": {
19    "extended_stats": {
20      "count": 50849,
21      "min": 48.0,
22      "max": 170000.0,
23      "avg": 7266.494778658381,
24      "sum": 3.69493993E8,
25      "sum_of_squares": 5.743992807611E12,
26      "variance": 6.015981900786375E7,
27      "variance_population": 6.015981900786375E7,
28      "variance_sampling": 6.0161002138350844E7,
29      "std_deviation": 7756.27610441143,
30      "std_deviation_population": 7756.27610441143,
31      "std_deviation_sampling": 7756.352373271269,
32      "std_deviation_bounds": {
33        "upper": 22779.04698748124,
34        "lower": -8246.057430164481,
35        "upper_population": 22779.04698748124,
36        "lower_population": -8246.057430164481,
37        "upper_sampling": 22779.19952520092,
38        "lower_sampling": -8246.209967884159
39      }
40    }
41  }
42 }
```

Cardinality clause - to get number of unique values in a field

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History Settings Help 200 - OK 270 ms

```
36 { "add": { "index": "cars_new", "alias": "cars" } }
37 }
38 }
39 GET /cars/_search
40 {
41   "size": 0,
42   "aggs": {
43     "total_price": { "sum": { "field": "priceUSD" } },
44     "avg_price": { "avg": { "field": "priceUSD" } },
45     "min_price": { "min": { "field": "priceUSD" } },
46     "max_price": { "max": { "field": "priceUSD" } }
47   }
48 }
49 }
50
51 GET /cars/_search
52 {
53   "size": 0,
54   "aggs": {
55     "value_count": { "value_count": { "field": "priceUSD" } },
56     "price_stats": { "stats": { "field": "priceUSD" } },
57     "extended_stats": {
58       "extended_stats": { "field": "priceUSD" }
59     }
60   }
61 }
62 }
63 GET /cars/_search
64 {
65   "size": 0,
66   "aggs": {
67     "unique_makes": {
68       "cardinality": {
69         "field": "make.keyword"
70       }
71     }
72   }
73 }
74 }
```

Click to send request

```
1- {
2   "took": 58,
3   "timed_out": false,
4   "_shards": {
5     "total": 1,
6     "successful": 1,
7     "skipped": 0,
8     "failed": 0
9   },
10  "hits": {
11    "total": {
12      "value": 10000,
13      "relation": "gte"
14    },
15    "max_score": null,
16    "hits": [ ]
17  },
18  "aggregations": {
19    "unique_makes": {
20      "value": 51
21    }
22  }
23 }
24 }
```

Terms buckets

Console Search Profiler Grok Debugger Painless Lab BETA

History Settings Help 200 - OK 268 ms

```
51 GET /cars/_search
52 {
53   "size": 0,
54   "aggs": {
55     "value_count": { "value_count": { "field": "priceUSD" } },
56     "price_stats": { "stats": { "field": "priceUSD" } },
57     "extended_stats": {
58       "extended_stats": { "field": "priceUSD" }
59     }
60   }
61 }
62
63 GET /cars/_search
64 {
65   "size": 0,
66   "aggs": {
67     "unique_makes": {
68       "cardinality": {
69         "field": "make.keyword"
70       }
71     }
72   }
73 }
74
75 GET /cars/_search
76 {
77   "size": 0,
78   "aggs": {
79     "models_agg": {
80       "terms": {
81         "field": "color.keyword",
82         "size": 20,
83         "order": {
84           "_key": "asc"
85         }
86       }
87     }
88   }
89 }
```

Click to send request

```
1 {
2   "took": 16,
3   "timed_out": false,
4   "_shards": {
5     "total": 1,
6     "successful": 1,
7     "skipped": 0,
8     "failed": 0
9   },
10  "hits": {
11    "total": {
12      "value": 10000,
13      "relation": "gte"
14    },
15    "max_score": null,
16    "hits": [ ]
17  },
18  "aggregations": {
19    "models_agg": {
20      "doc_count_error_upper_bound": 0,
21      "sum_other_doc_count": 0,
22      "buckets": [
23        {
24          "key": "black",
25          "doc_count": 11256
26        },
27        {
28          "key": "blue",
29          "doc_count": 7420
30        },
31        {
32          "key": "brown",
33          "doc_count": 1186
34        },
35        {
36          "key": "burgundy",
37          "doc_count": 1842
38        },
39      ]
40    }
41  }
42 }
```

Nested buckets, e.g. Year -> Fuel Type -> Price Stats

Console Search Profiler Grok Debugger Painless Lab BETA

History Settings Help 200 - OK 316 ms

```
71 }
72 }
73 }
74
75 GET /cars/_search
76 {
77   "size": 0,
78   "aggs": {
79     "models_agg": {
80       "terms": {
81         "field": "color.keyword",
82         "size": 20,
83         "order": {
84           "_key": "asc"
85         }
86       }
87     }
88   }
89 }
90
91 GET /cars/_search
92 {
93   "size": 0,
94   "aggs": {
95     "by_year": {
96       "terms": { "field": "year", "size": 100, "order": { "_key": "asc" } },
97       "aggs": {
98         "by_fuel": {
99           "terms": { "field": "fuel_type.keyword", "size": 10 },
100          "aggs": {
101            "price_stats": { "stats": { "field": "priceUSD" } }
102          }
103        }
104      }
105    }
106  }
107 }
108 }
```

Click to send request

```
1870 {
1871   {
1872     "key": "1546300800000",
1873     "key_as_string": "2019",
1874     "doc_count": 183,
1875     "by_fuel": {
1876       "doc_count_error_upper_bound": 0,
1877       "sum_other_doc_count": 0,
1878       "buckets": [
1879         {
1880           "key": "petrol",
1881           "doc_count": 144,
1882           "price_stats": {
1883             "count": 144,
1884             "min": 5500.0,
1885             "max": 122290.0,
1886             "avg": 27220.381944444445,
1887             "sum": 3919735.0
1888           }
1889         },
1890         {
1891           "key": "diesel",
1892           "doc_count": 39,
1893           "price_stats": {
1894             "count": 39,
1895             "min": 19722.0,
1896             "max": 111690.0,
1897             "avg": 52050.179487179485,
1898             "sum": 2029957.0
1899           }
1900         }
1901       ]
1902     }
1903   }
1904 }
1905 }
1906 }
1907 }
1908 }
```

Minimum document count to filter sparse buckets

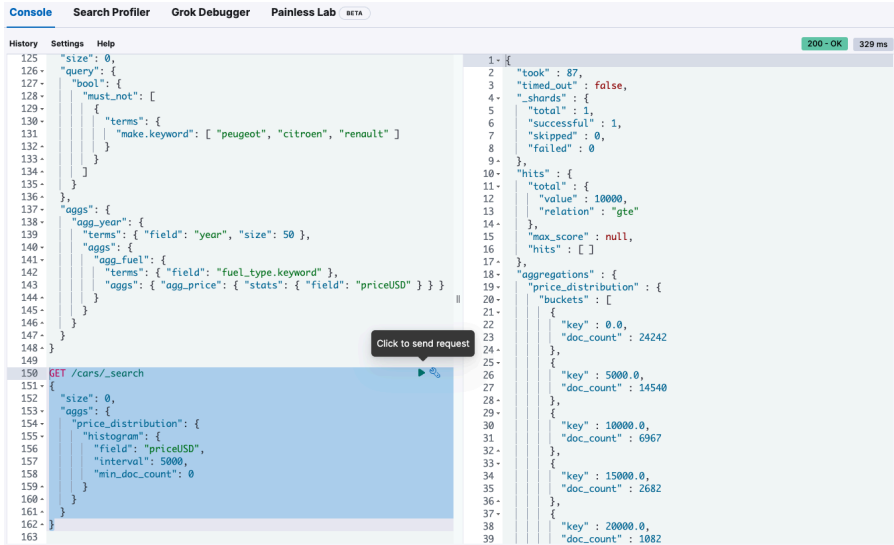
ConsoleSearch ProfilerGrok DebuggerPainless LabBETA

HistorySettingsHelp200 - OK290 ms

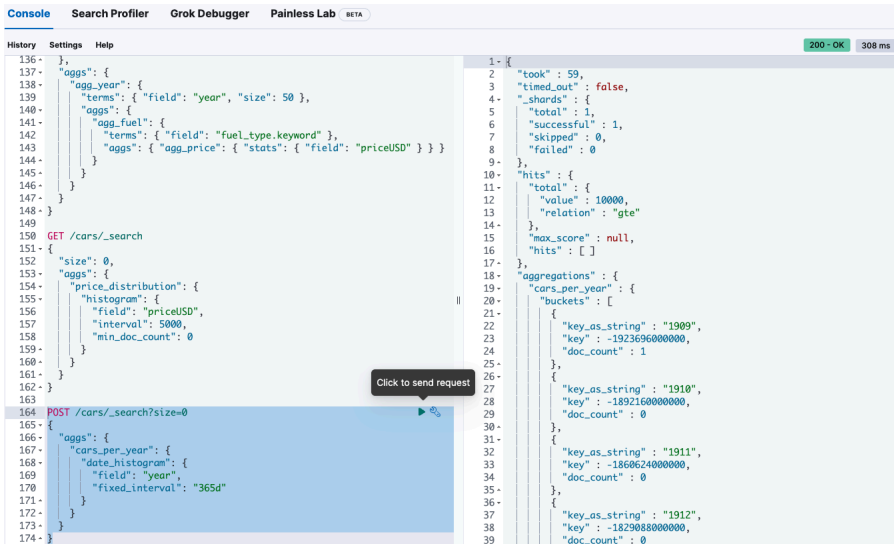
84- "_key": "asc"
85- }
86- }
87- }
88- }
89- }
90- }
91- GET /cars/_search
92- {
93- "size": 0,
94- "aggs": {
95- "by_year": {
96- "terms": { "field": "year", "size": 100, "order": { "_key"
97- : "asc" } },
98- "by_fuel": {
99- "terms": { "field": "fuel_type.keyword", "size": 10 },
100- "aggs": {
101- "price_stats": { "stats": { "field": "priceUSD" } }
102- }
103- }
104- }
105- }
106- }
107- }
108- }
109- GET /cars/_search
110- {
111- "size": 0,
112- "aggs": {
113- "top_makes": {
114- "terms": {
115- "field": "make.keyword",
116- "size": 30,
117- "min_doc_count": 1000
118- }
119- }
120- }
121- }
122- }
123- }
124- }
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Histograms:

Numeric histogram



Yearly date histogram



Missing-Value Handling & Update

ConsoleSearch ProfilerGrok DebuggerPainless LabBETA

HistorySettingsHelp

158-
159-
160-
161-
162-
163-
164- POST /cars/_search?size=0
165-
166- {
167- "aggs": {
168- "cars_per_year": {
169- "date_histogram": {
170- "field": "year",
171- "fixed_interval": "365d"
172- }
173- }
174- }
175- }
176- POST cars/_update_by_query
177- {
178- "script": {
179- "source": "ctx._source.make = null",
180- "lang": "painless"
181- },
182- "query": {
183- "bool": {
184- "must": [
185- {
186- "term": {
187- "make.keyword": {
188- "value": "mazda"
189- }
190- }
191-]
192- }
193- }
194- }
195- }
196- }

Click to send request

200 - OK1659 ms

1- {
2- "took" : 1414,
3- "timed_out" : false,
4- "total" : 1837,
5- "updated" : 1837,
6- "deleted" : 0,
7- "batches" : 2,
8- "version_conflicts" : 0,
9- "noops" : 0,
10- "retries" : {
11- "bulk" : 0,
12- "search" : 0
13- },
14- "throttled_millis" : 0,
15- "requests_per_second" : -1.0,
16- "throttled_until_millis" : 0,
17- "failures" : []
18- }
19- }

Filter out nulls

ConsoleSearch ProfilerGrok DebuggerPainless LabBETA

HistorySettingsHelp

165- {
166- "aggs": {
167- "cars_per_year": {
168- "date_histogram": {
169- "field": "year",
170- "fixed_interval": "365d"
171- }
172- }
173- }
174- }
175- POST cars/_update_by_query
176- {
177- "script": {
178- "source": "ctx._source.make = null",
179- "lang": "painless"
180- },
181- "query": {
182- "bool": {
183- "must": [
184- {
185- "term": {
186- "make.keyword": {
187- "value": "mazda"
188- }
189- }
190-]
191- }
192- }
193- }
194- }
195- }
196- }
197- GET cars/_search
198- {
199- "query": {
200- "exists": { "field": "make" }
201- }
202- }
203- }

Click to send request

200 - OK261 ms

1- {
2- "took" : 11,
3- "timed_out" : false,
4- "_shards" : {
5- "total" : 1,
6- "successful" : 1,
7- "skipped" : 0,
8- "failed" : 0
9- },
10- "hits" : {
11- "total" : {
12- "value" : 18000,
13- "relation" : "gte"
14- },
15- "max_score" : 1.0,
16- "hits" : [
17- {
18- "_index" : "cars_new",
19- "_id" : "Vfy8PZgBG17L8YkGk_TP",
20- "_score" : 1.0,
21- "_source" : {
22- "make" : "bmw",
23- "priceUSD" : "5400",
24- "year" : "2001",
25- "mileage(kilometers)" : "405000.0",
26- "fuel.type" : "petrol",
27- "volume(cm3)" : "3000.0",
28- "color" : "black",
29- "transmission" : "auto",
30- "segment" : "E"
31- }
32- },
33- {
34- "_index" : "cars_new",
35- "_id" : "Vvy8PZgBG17L8YkGk_TP",
36- "_score" : 1.0,
37- "_source" : {
38- "make" : "bmw",
39- "priceUSD" : "14500",
40- }
41- }
42-]
43- }
44- }

Aggregate on missing

ConsoleSearch ProfilerGrok DebuggerPainless LabBETA

HistorySettingsHelp

178- {
179- "script": {
180- "source": "ctx._source.make = null",
181- "lang": "painless"
182- },
183- "query": {
184- "bool": {
185- "must": [
186- {
187- "term": {
188- "make.keyword": {
189- "value": "mazda"
190- }
191- }
192-]
193- }
194- }
195- }
196- }
197- GET cars/_search
198- {
199- "query": {
200- "exists": { "field": "make" }
201- }
202- }
203- }
204- GET /cars/_search
205- {
206- "size": 0,
207- "aggs": {
208- "missing_makes": {
209- "missing": { "field": "make.keyword" },
210- "aggs": {
211- "by_color": {
212- "terms": { "field": "color.keyword", "size": 30 }
213- }
214- }
215- }
216- }
217- }

Click to send request

200 - OK263 ms

1- {
2- "took" : 25,
3- "timed_out" : false,
4- "_shards" : {
5- "total" : 1,
6- "successful" : 1,
7- "skipped" : 0,
8- "failed" : 0
9- },
10- "hits" : {
11- "total" : {
12- "value" : 18000,
13- "relation" : "gte"
14- },
15- "max_score" : null,
16- "hits" : []
17- },
18- "aggregations" : {
19- "missing_makes": {
20- "doc_count" : 1837,
21- "by_color" : {
22- "doc_count_error_upper_bound" : 0,
23- "sum_other_doc_count" : 0,
24- "buckets" : [
25- {
26- "key" : "silver",
27- "doc_count" : 368
28- },
29- {
30- "key" : "blue",
31- "doc_count" : 272
32- },
33- {
34- "key" : "black",
35- "doc_count" : 248
36- },
37- {
38- "key" : "gray",
39- "doc_count" : 212
40- }
41-]
42- }
43- }
44- }
45- }